SMILE: A Hybrid Memory-Driven Dialogue Manager for Social Companion Robots

Lorenzo D'Errico[2], Salvatore M. Anzalone[3,4], and Mariacarla Staffa[1]

***Abstract*—This paper presents SMILE (Social Memory-Integrated Language Environment), a memory-augmented dialogue system designed for social companion robots. Current LLM-based conversational agents are stateless by design, creating a fundamental tension between their apparent linguistic intelligence and the persistent, structured memory required for long-term social interaction — a limitation particularly critical in assistive contexts involving older adults or individuals with cognitive impairments. SMILE addresses this gap through a controlled, locally-run architecture combining real-time face recognition, offline speech processing, and a three-layer memory structure comprising working, semantic, and episodic memory, in which the LLM serves exclusively as a linguistic realizer while memory retrieval and dialogue control remain deterministic. Rather than presenting experimental results, this work describes the system's design and architectural rationale, discussing key trade-offs and open challenges at the intersection of human memory models, assistive robotics, and LLM-based dialogue systems.**

***Index Terms*—Social robotics, conversational agents, memory-augmented dialogue, large language models, assistive robotics, human–robot interaction**

I. Introduction

Social robots are progressively moving from controlled laboratory environments to real-world settings such as elderly care, cognitive rehabilitation, education, and everyday assistance. In these contexts, robots are not merely expected to execute isolated tasks, but to sustain long-term, meaningful relationships with users, involving continuity over time, personalization, and the ability to remember previous encounters. Despite significant progress in perception, navigation, and speech technologies, most deployed social robots still operate with very limited memory: they can recognise a face or transcribe speech, but rarely remember who they are talking to in a structured way, nor relate current events to past experiences with the same person. As a consequence, interactions often feel shallow and repetitive, and users may perceive the robot as less socially credible than its physical embodiment suggests. Large Language Models (LLMs) have recently transformed natural language interaction, but are by design stateless: they only "remember" what is explicitly provided in the input prompt. This creates a tension between apparent conversational intelligence and the lack of persistent, structured memory required for long-term social interaction. In this work we present SMILE (Social Memory-Integrated Language Environment), a memory-augmented dialogue system for social companion robots. Rather than proposing a fully validated solution, SMILE represents an architectural exploration of how persistent identity, multi-layer memory, and contextually appropriate recall might be integrated into a controllable, locally-run pipeline combining ASR, TTS, face-based user identification, and an LLM hosted via Ollama. The system maintains working memory of recent utterances, long-term semantic memory of stable user attributes, and a growing episodic memory of past experiences, with the goal of sustaining coherent, personalised conversations across sessions, including for research and clinical purposes involving older adults or individuals with cognitive impairments. The remainder of this paper briefly reviews the relevant background in Sec.II, then presents SMILE through a system overview Sec.III, a detailed description of the memory architecture in Sec.IV, the dialogue management layer in Sec.V, and the technical implementation in Sec.VI, before closing with a critical discussion of limitations and open challenges.

II. Background and Related Work

The development of conversational social robots lies at the intersection of three major research domains: cognitive robotics, memory modelling, and dialogue systems. Within cognitive psychology, memory is commonly conceptualised as a set of interacting subsystems, including working memory for short-term active processing, semantic memory for consolidated factual knowledge, and episodic memory for autobiographical experiences grounded in spatial and temporal context [1, 2]. Beyond this classical taxonomy, recent perspectives emphasise the constructive, context-dependent, and socially embedded nature of memory processes, particularly in interactive settings where memory contributes to identity and relational continuity [3].

The taxonomy illustrated in Fig. 1 has strongly influenced cognitive robotics architectures such as Soar [4] and CRAM [5], which implement layered memory systems enabling reasoning grounded in past experience. A central insight emerging from this tradition is that memory operates as a context-sensitive trigger: it is selectively activated by cues rather than remaining continuously accessible [3]. In human-robot interaction (HRI), this perspective has been extended toward the notion that memory is also fundamental for enabling person-

The work was supported by "RESTART - Robot Enhanced Social abilities based on Theory of mind for Acceptance of Robot in assistive Treatments" (CUP: I53D23003780001), funded by the MIUR with D.D. no.861 under the PNRR and by Next Generation EU.

[1]University of Naples Parthenope, Naples, Italy. mariacarla.staffa@uniparthenope.it
[2]University of Naples Federico II, Naples, Italy. lorenzo.derrico@unina.it
[3]Laboratoire CHArt, Université Paris 8, Saint Denis, France. sanzalone@univ-paris8.fr
[4]LUTIN, Cité des Sciences et de l'Industrie, Paris, France.

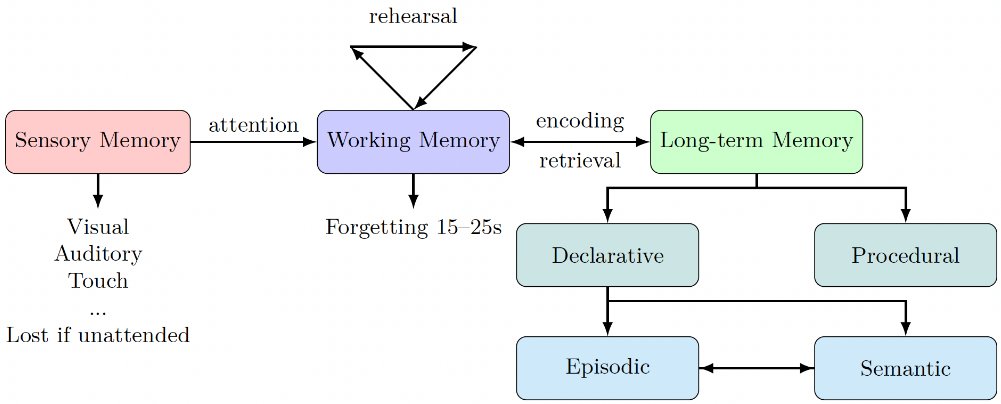

Fig. 1. Taxonomy of human memory systems, illustrating the flow from sensory input to working memory and the division of long-term memory into declarative (episodic and semantic) and procedural components.

alisation, user modelling, and perceived identity consistency over time. In particular, prior work has shown that adapting robot behaviour to user personality traits and social identity significantly affects engagement and interaction quality [6]. Similarly, architectures designed for long-term interaction highlight the importance of maintaining persistent user representations and behavioural adaptation across sessions. More recently, Large Language Models (LLMs) have significantly advanced the fluency and flexibility of dialogue systems. However, these models are inherently stateless: they do not retain information across interactions unless it is explicitly reintroduced within the prompt. As a consequence, equipping LLM-based agents with persistent memory requires external mechanisms such as context management strategies, retrieval-augmented generation, or summarisation pipelines [7]. In practice, most LLM-driven social robots rely on persona-based prompts or raw dialogue histories. While simple to implement, these approaches scale poorly over long-term interactions and often lead to inconsistencies that users—particularly older adults—readily detect [8]. This limitation directly impacts the perception of coherence and trust, which are known to depend on stable behavioural and relational patterns in HRI [9]. Conversely, established cognitive architectures provide richer and more structured memory representations, but their symbolic complexity and computational demands pose challenges for real-time integration with LLMs, especially on resource-constrained domestic hardware. SMILE is positioned within this gap. It proposes a hybrid architecture that combines the linguistic fluency of LLMs with deterministic and structured memory mechanisms, enabling persistent identity, longitudinal user modelling, and episodic recall. This design aligns with recent research trends in HRI that emphasise lifelong learning, adaptive interaction, and the integration of cognitive and affective models to sustain long-term engagement [10, 6].

III. System Overview

SMILE (Social Memory-Integrated Language Environment) is designed as a hybrid architecture combining multimodal perception, explicit memory structures, and controlled language generation. The core design principle is that memory, context handling, and interaction flow are externalised from the LLM and managed deterministically through code: the LLM acts exclusively as a linguistic realiser, transforming structured prompts into natural utterances. This separation prevents hallucinations, ensures predictable behaviour, and maintains the trust and coherence essential in assistive robotics contexts. As illustrated in Figure 2, the system is organised into four sequential layers. The Perception Layer processes video and audio input: faces are detected and encoded as embeddings via MTCNN [11] and FaceNet [12], enabling persistent user identification across sessions without requiring explicit re-introduction. The Conversation Layer manages the full dialogue pipeline — a finite-state machine (FSM) governs interaction flow, a dialogue manager constructs structured prompts, a LLaMA-based LLM running locally via Ollama generates responses, and a TTS engine renders them as audio output. The Memory Layer maintains three complementary stores: working memory of recent dialogue turns, historical summaries, and episodic storage of discrete past experiences. Finally, the Profile Layer persists user information as structured JSON files, updated after each session through a consolidation pipeline that extracts semantic attributes and episodic events from working memory.

A. Multimodal Perception

The vision module continuously samples camera frames, detects faces via MTCNN, and encodes them using FaceNet embeddings compared via cosine similarity against stored profiles. Recognised users have their profiles loaded immediately; unknown users trigger a name acquisition sequence and profile creation. On the audio side, a tuned voice activity detector (VAD) monitors RMS energy and silence durations to regulate turn-taking. Speech is transcribed using the Vosk offline ASR engine and normalised before entering the dialogue manager, reducing ASR noise propagation to downstream components.

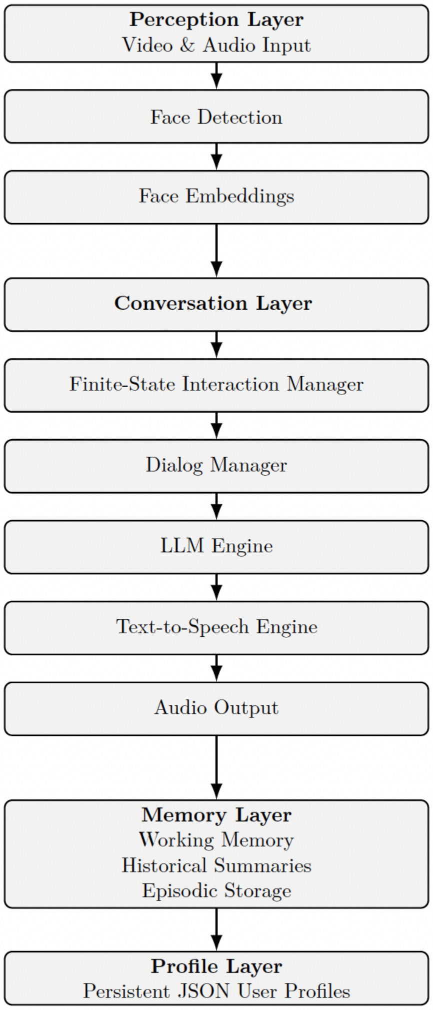

Fig. 2. SMILE overall architecture from perception to persistent memory.

Offline processing ensures privacy and low latency — critical for home deployment with vulnerable users.

IV. Memory Architecture

SMILE implements an explicit, structured memory system inspired by the cognitive taxonomy discussed in Section II, adapted to the constraints of LLM-based dialogue and assistive robotics, where simplicity, transparency, and predictability take precedence over computational sophistication. Three principles guide the design: strict separation of memory types, with distinct access patterns and persistence characteristics for each layer; controlled access, where memory is never queried directly by the LLM but mediated entirely by the dialogue manager under explicit rules; and verifiable grounding, ensuring all stored information originates from the user's own words rather than model inference.

Fig. 3 shows the three level memory system design that is described in detail below:

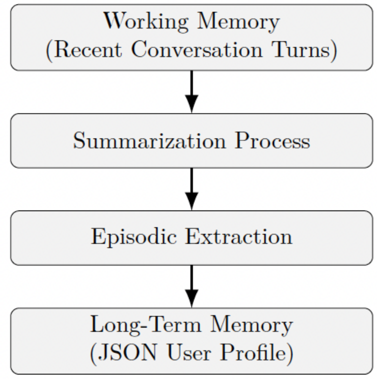

Fig. 3. Working memory is consolidated through LLM-based summarization, extracting both stable semantic attributes and discrete episodic events that are stored in the persistent user profile.

A. Working Memory

Working memory maintains the last seven conversational turns as structured (user, robot) message pairs, supporting local coherence, pronoun resolution, and multi-turn continuity. The buffer resets at session end: information is retained long-term only if the post-session consolidation pipeline identifies stable attributes or salient episodic events. Strict control over the buffer also prevents the LLM from reintroducing irrelevant details from earlier turns — a common failure mode in unconstrained LLM dialogue.

B. Semantic Memory

Semantic memory stores stable user attributes — estimated age, occupation, interests, personality descriptors, long-term goals — as a structured JSON profile. It is updated conservatively, integrating only attributes explicitly grounded in user statements. The dialogue manager selects which attributes to inject into the prompt based on the current conversational topic, avoiding intrusive or contextually irrelevant references to stored personal information.

C. Episodic Memory

Episodic memory stores personal events mentioned by the user, each represented as a structured record with a title, temporal reference, concise summary, and semantic tags. Retrieval is rule-based and cue-dependent: episodes are included in the LLM prompt only when the user's current utterance overlaps with an episode's tags or semantically related keywords, mirroring the triggered nature of human episodic recall [3]. This deterministic pipeline ensures memory usage remains explainable and auditable.

D. Coordination and Consolidation

When a new utterance is received, the dialogue manager updates working memory, queries semantic memory for topic-relevant attributes, searches episodic memory via tag-matching, and integrates only the selected fragments into the final prompt. At session end, the consolidation pipeline uses a carefully constrained LLM prompt to extract semantic attributes and episodic events from the working memory buffer, explicitly instructing the model to reject any inference not directly stated by the user. The resulting updates are merged into the persistent user profile, with conflicts resolved by prioritising the most recent information.

V. Dialogue Management

The dialogue manager is the core component governing SMILE's moment-to-moment behaviour. Rather than delegating conversational control to the LLM, SMILE adopts an explicit, deterministic strategy that ensures predictability, safety, and interpretability — essential properties for assistive robotics contexts.

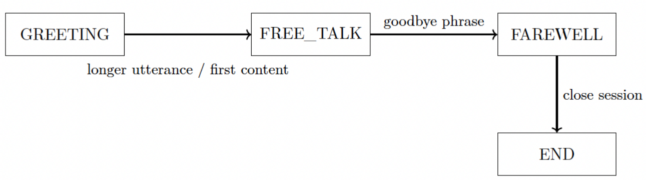

Fig. 4. Finite-state machine governing conversational flow in SMILE.

A. Finite-State Machine

As illustrated in Figure 4, the dialogue manager operates as a finite-state machine (FSM) with three states. GREETING is triggered upon user recognition; memory access is minimal to avoid overwhelming the user during initial contact. FREE_TALK is the main conversational state, with full memory access, topic-following, and behavioural constraints active. FAREWELL is entered when closing intent is detected via keyword matching; the robot produces a brief goodbye without prolonging the interaction. Transitions are deterministic: GREETING advances to FREE_TALK as soon as the user produces substantive content; FREE_TALK transitions to FAREWELL on explicit closing expressions. Upon session termination, the consolidation pipeline is triggered.

B. Behavioural Constraints and Prompt Engineering

Within each state, the dialogue manager enforces behavioural constraints through structured prompt construction. The prompt acts as a behavioural contract for the LLM, embedding rules such as: no repeated greetings outside GREETING; replies limited to 2–3 sentences to avoid cognitive overload; no unsolicited topic changes; no emotional assumptions not grounded in explicit user input; confirmation requests when utterances are unclear or potentially misrecognised; and selective, context-driven memory use as determined by the manager. These constraints prevent common failure modes of generative systems, verbosity, hallucinated emotional interpretations, and unwanted topic shifts.

C. Memory Retrieval and Prompt Construction

For each user utterance, the dialogue manager executes the following pipeline: it identifies the current FSM state and applies state-specific rules; retrieves relevant semantic attributes via topic matching, the last seven working memory turns, and any episodic memories whose tags overlap with the current utterance; constructs the full structured prompt; queries the local LLM via Ollama; and post-processes the response to enforce conciseness, remove unwanted greetings, and filter repetitive content. The LLM never retrieves memory independently — all retrieval is deterministic and mediated by code, ensuring explainability and auditability of every memory access.

D. Error Handling

Real-world speech recognition introduces significant uncertainty. The dialogue manager handles this conservatively: unclear utterances trigger explicit confirmation requests; suspected ASR errors are ignored unless the user repeats the same content; working memory context is used to disambiguate noisy input; and the system refuses to assume facts not present in memory or explicitly stated by the user. This conservative stance prioritises reliability over fluency — appropriate for users where misinterpretations may cause confusion or distress.

E. Post-Conversation Consolidation

Upon entering FAREWELL, SMILE triggers an offline consolidation phase. A specialised summarisation prompt processes the working memory buffer, extracting new stable attributes for semantic memory, identifying discrete episodic events for long-term storage, updating conversation summaries and retrieval tags, and explicitly rejecting any attribute not directly stated by the user. This separation of online interaction from reflective memory encoding ensures memory quality and prevents the accumulation of erroneous information across sessions.

VI. System Implementation

SMILE is implemented entirely in Python and designed to run on commodity hardware with GPU acceleration, enabling real-time multimodal interaction suitable for assistive robotics deployment. The architecture follows a layered, modular design articulated in five functional layers: vision (face detection, tracking, and embedding-based identification), audio (ASR and TTS), dialogue (finite-state machine), memory (working, semantic, and episodic subsystems), and language (local LLM inference via Ollama). This separation of concerns ensures modularity and transparent behavioral control—critical properties for systems deployed with vulnerable populations. The complete interaction cycle is depicted in Figure 5.

a) Multimodal Perception.: The vision module processes webcam frames using MTCNN [13] and extracts 128-dimensional embeddings via FaceNet [12]; cosine similarity above 0.6 triggers profile retrieval, while unrecognized faces initiate a registration dialogue. Face processing runs

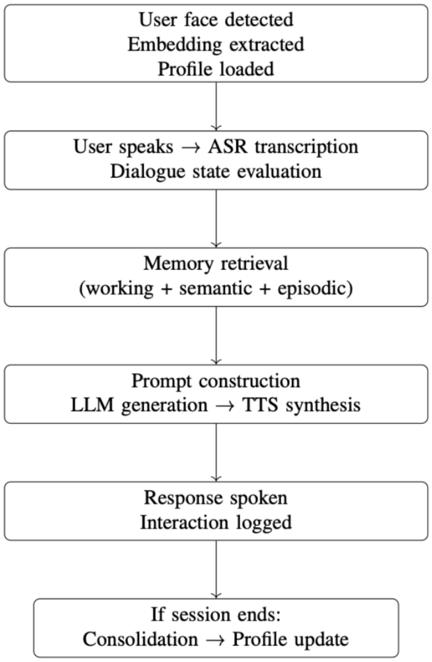

Fig. 5. Complete interaction cycle in SMILE, from perception through response generation to memory consolidation.

asynchronously to avoid blocking the main loop [14]. Audio capture employs continuous microphone streaming with adaptive RMS-based silence detection for turn-taking [15]; speech is transcribed offline via Vosk (`vosk-model-it-0.22`), preserving privacy by avoiding cloud dependencies. TTS synthesis via `pyttsx3` executes asynchronously, eliminating perceptible response delays.

b) Dialogue Management and LLM Integration.: The dialogue manager implements a three-state FSM (`GREETING`, `FREE_TALK`, `FAREWELL`) with transitions governed by explicit lexical patterns rather than LLM inference, ensuring deterministic and explainable behavior. At each turn, a structured prompt is assembled from:

- current dialogue state and behavioral constraints;
- filtered semantic memory attributes (age, occupation, interests);
- contextually matched episodic memories via tag-based retrieval;
- recent working memory trace (last 7 turns) and temporal grounding.

Meta-instructions explicitly prohibit repeated greetings, unsolicited topic changes, and verbose responses—following best practices for controlled generation in assistive social robotics [8]. Response generation is handled locally by `llama3` via Ollama, avoiding latency, cost, and privacy concerns of cloud-based APIs.

c) Memory Persistence, Consolidation, and Transparency.: User profiles are stored as structured JSON files containing semantic attributes (name, age range, interests, personality descriptors) and an episodic memory array (titled events with timestamps, summaries, and retrieval tags). Working memory resides solely in RAM and resets after each session [16]. At session termination, a consolidation module processes the working memory buffer to extract:

- new or refined semantic attributes explicitly stated by the user;
- discrete episodic events suitable for long-term storage;
- an updated conversational summary for the session.

Strict grounding instructions ensure only information directly supported by user utterances is retained, preventing hallucinated attributes from contaminating the profile. All interactions are logged in human-readable JSON, capturing utterances, responses, memory updates, and state transitions—supporting post-hoc analysis, longitudinal studies, clinical validation, and full reproducibility [17].

VII. Discussion and Limitations

SMILE demonstrates that combining structured memory with LLM-based dialogue generation offers a viable pathway for personalised, explainable, and locally deployable social robotics. The current work presents the system's architecture and theoretical grounding; no formal user validation has yet been conducted. The design process surfaced several honest trade-offs. SMILE deliberately prioritises transparency over flexibility: the FSM and prompt-constrained LLM ensure safe, auditable behaviour, but prompt instructions remain soft constraints vulnerable to conversational drift or ASR noise. The semantic-episodic distinction, while architecturally clean, introduces practical ambiguities, as user utterances often blend stable preferences with time-bound experiences, and consolidation heuristics can produce inconsistent classifications across sessions. At the retrieval level, tag-based matching prevents inappropriate recall but cannot respond to implicit cues or prioritise memories by recency and emotional salience — more flexible approaches such as embedding-based retrieval would enrich recall but introduce opacity incompatible with assistive deployment. At the perceptual level, ASR errors propagate directly into memory consolidation and are difficult to detect automatically, while face recognition degrades with appearance changes or variable lighting. On the ethical side, SMILE currently provides no mechanisms for users to view, edit, or delete their stored profiles, conflicting with GDPR principles; sensitive disclosures and the risk of emotional overreliance in vulnerable users will require dedicated safeguards in future iterations. Evaluating memory-augmented conversational systems for social robotics is itself a complex methodological challenge: unlike task-oriented dialogue, companion robots require assessment across technical performance, memory coherence, and user experience simultaneously, with traditional automatic metrics correlating poorly with dialogue quality. A rigorous evaluation of SMILE — including controlled sessions, longitudinal studies, and eventual deployment with vulnerable populations, falls outside the scope of this work and will be the subject of future studies.

VIII. Conclusion

SMILE demonstrates that a carefully designed hybrid architecture, combining deterministic dialogue control, structured memory persistence, and LLM-based language generation, provides a practical foundation for memory-augmented social robotics. The system addresses key limitations of purely end-to-end LLM approaches by integrating face-based user identification, a three-layer cognitively-inspired memory model, and FSM-governed conversational flow, all deployable on commodity hardware without cloud dependencies. The key contributions can be summarized as follows: persistent cross-session memory with automatic consolidation; deterministic, auditable dialogue control; controlled and explainable memory retrieval; and a fully local deployment pipeline that preserves user privacy. These properties are not merely technical conveniences, they are prerequisites for responsible deployment in assistive contexts involving vulnerable populations. Acknowledged limitations, prompt compliance fragility, retrieval rigidity, semantic, episodic boundary ambiguity, and ASR dependency, reflect broader open challenges in memory-aware conversational agents rather than isolated implementation flaws. Addressing them will require hybrid retrieval mechanisms that balance transparency with cognitive flexibility, adaptive consolidation policies, and forgetting strategies for long-running profiles. Future extensions will prioritize emotion-aware memory tagging, richer dialogue policies, and user-facing profile transparency tools consistent with data protection requirements. The proposed evaluation framework establishes a systematic pathway from technical validation to longitudinal deployment. Ultimately, SMILE serves as both a functioning prototype and an experimental platform for exploring the intersection of cognitive memory models, LLM capabilities, and embodied social interaction—a step toward social robots capable of remembering, learning, and adapting to individual users over time.

Code and Data Availability

The complete implementation of SMILE is publicly available under the MIT License at following github repository. SMILE is an open research project. Contributions, bug reports, and feature requests are welcome via GitHub Issues or Pull Requests. Researchers deploying SMILE in their own studies are encouraged to share anonymized results and implementation experiences to advance the broader community's understanding of memory-augmented social robotics.

References

[1] Richard C Atkinson and Richard M Shiffrin. Human memory: A proposed system and its control processes. *Psychology of Learning and Motivation*, 2:89–105, 1968.

[2] Martin A Conway and Christopher W Pleydell-Pearce. The construct of autobiographical memory: theoretical perspectives. *Psychological Bulletin*, 127(3):261, 2001.

[3] Endel Tulving. Episodic memory: From mind to brain. *Annual Review of Psychology*, 53(1):1–25, 2002.

[4] John E Laird, Allen Newell, and Paul S Rosenbloom. Soar: An architecture for general intelligence. *Artificial Intelligence*, 33(1):1–64, 1987.

[5] Michael Beetz, Lorenz Mösenlechner, and Moritz Tenorth. Cram — a cognitive robot abstract machine for everyday manipulation in human environments. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1012–1017. IEEE, 2010.

[6] Mariacarla Staffa, Lorenzo D'Errico, and Antonio Maratea. Influence of social identity and personality traits in human-robot interactions. *Robotics*, 2024.

[7] Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. A survey on the memory mechanism of large language model-based agents. *ACM Transactions on Information Systems*, 43(6):1–47, 2025.

[8] Bahar Irfan, Sanna Kuoppamäki, Aida Hosseini, and Gabriel Skantze. Between reality and delusion: challenges of applying large language models to companion robots for open-domain dialogues with older adults. *Autonomous Robots*, 49(1):9, 2025.

[9] Silvia Rossi, Mariacarla Staffa, et al. The role of personality factors and empathy in the acceptance and performance of a social robot. *Robotics*, 2020.

[10] Mariacarla Staffa et al. Classifying human emotions in hri: applying global optimization model to eeg brain signals. *Frontiers in Neurorobotics*, 2023.

[11] Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. Joint face detection and alignment using multitask cascaded convolutional networks. In *IEEE Signal Processing Letters*, 2016.

[12] Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823, 2015.

[13] Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. Joint face detection and alignment using multitask cascaded convolutional networks. In *IEEE Signal Processing Letters*, pages 1499–1503, 2016.

[14] Dan Bohus and Alexander I. Rudnicky. The ravenclaw dialog management framework: Architecture and systems. In *Computer Speech & Language*, volume 23, pages 332–361, 2009.

[15] Antoine Raux and Maxine Eskenazi. Optimizing endpointing thresholds using dialogue features in a spoken dialogue system. In *Proc. of the 9th SIGdial Workshop on Discourse and Dialogue*, pages 1–10, Columbus, Ohio, 2008. Association for Computational Linguistics.

[16] Alan Baddeley. Working memory: Theories, models, and controversies. *Annual Review of Psychology*, 63:1–29, 2012.

[17] Fabian Peller-Konrad, Rainer Kartmann, Christian R. G. Dreher, Andre Meixner, Fabian Reister, Markus Grotz, and Tamim Asfour. A memory system of a robot cognitive architecture and its implementation in armarx. *Robotics and Autonomous Systems*, 164:104415, 2023.